

Mapping and Modeling Mangrove Loss in Palawan, Philippines: A Biogeographic Study of Environmental Drivers

J.D. Casisirano^{1,2*}

¹Department of Geodetic Engineering, University of the Philippines, Diliman, Quezon City, Philippines
(jarencacasisirano@gmail.com)

²Training Center for Applied Geodesy and Photogrammetry, University of the Philippines, Diliman, Quezon City, Philippines

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Abstract

Mangrove ecosystems provide essential ecological services, yet continue to face pressure from environmental stressors and land-use change. This study analyzed the spatial and temporal dynamics of mangrove loss in Palawan, Philippines, over a 24-year period (1996–2020), using remote sensing-based change detection, regression modeling, and spatial analysis. Mangrove extent data from the Global Mangrove Watch were processed to detect gain and loss across three intervals: 1996–2010, 2010–2015, and 2015–2020. Results showed that the most substantial decline occurred in the first interval, while more recent years exhibited a slowdown in loss and near-equilibrium with gain. Regression analysis revealed elevation and rainfall as consistent environmental drivers of mangrove loss, while spatial modeling through Geographically Weighted Regression (GWR) highlighted variation in these effects across the province. The integration of pixel-level predictors and local modeling confirmed that environmental drivers do not operate uniformly, reflecting biogeographic variability in mangrove dynamics. These findings underscore the need for localized conservation strategies and demonstrate that mangrove loss is shaped by both stable geographic gradients and temporally dynamic environmental influences.

1. Introduction

1.1 Background of the Study

Mangrove forests are wetlands found along the tropical and subtropical coast, specifically in the intertidal zone where land meets the sea, often exposed to the rising and falling of tides. It is one of the most unique, productive, and ecologically important ecosystems that can thrive in extremely harsh environmental conditions (Nizam et al., 2002), unlike other forest ecosystems. Mangrove's uniqueness allows them to adapt in areas with high salinity, high temperature, extreme tides, strong winds, muddy and anaerobic soils (Kathiresan & Bingham, 2001). With their unique adaptive mechanism and resiliency to harsh environmental conditions, mangroves had occupied the once unproductive coastlines and turned it into one of the most productive ecosystems.

Mangroves provide a wide range of ecosystem services (Bimrah et al., 2022; Getzner & Islam, 2020) including provisioning services, regulating services, cultural services, and supporting services. Provisioning services of mangroves include raw materials, and food provisioning. Along this, it also offers regulating services by regulating climate through its carbon sequestration ability, and protection of coastal areas from natural hazards. This ecosystem also offers cultural services through their aesthetic value and opportunities for both recreation and tourism, while also offering supporting services through maintenance and protection of biodiversity (Mitra, 2020). These services provided by the mangrove ecosystem contribute to the improvement of the well-being of humankind.

The Philippines, being a megadiverse country, is home to different flora and fauna, including a variety of mangrove species. According to Primavera et al. (2004), the country itself

holds at least 35 to 40 species of mangroves out of 70 species worldwide. However, over the decades, these ecosystems have faced considerable threats, leading to widespread degradation and loss. Hence, understanding the spatial distribution of mangroves and the environmental factors that influence their extent is crucial in assessing the biogeography of mangrove loss and informing conservation strategies.

As mangrove forests decline at an alarming rate, it has become a very important concern for conservationists to monitor the status and condition of mangroves. On site observation and field inventory have been the traditional way of monitoring mangrove forests. However, this is often time-consuming, resource-intensive, and limited in spatial scope, especially in remote or inaccessible areas. With remote sensing, a robust tool that enables the widespread detection and mapping of mangroves, it is now possible to monitor and map a large extent of mangrove forest at much faster rate and lower cost than the traditional way (Pillodar et al., 2023). Using data from remote sensing, spatial patterns including the location, distribution, and density of mangrove forest can be observed, making it possible to track the historical changes within the forests. Furthermore, remote sensing also allows the observation of the ecological trends of mangrove forests through identification of the patterns and extent of their contraction and expansion.

Assessing the impact and influence of both environmental and anthropogenic variables in mangroves is also necessary to appropriately construct evidence-based and scientific based conservation efforts. Often, a regression modeling is used to observe the effects of variables on one another (Burton, 2021). Logistic regression provides a global assessment of how environmental variables—such as rainfall, temperature, and elevation—influence the likelihood of mangrove loss. Unlike linear models, logistic regression is suited for binary outcomes,

making it appropriate for analyzing the presence or absence of mangrove loss at specific locations. However, environmental influences are not spatially uniform. To account for the spatial heterogeneity in the geographic distribution of mangrove dynamics, Geographically Weighted Regression (GWR) can be employed to examine how the strength and direction of these relationships vary across locations (Rahmila et al., 2025). Together, these methods offer a comprehensive framework for evaluating both the spatio-temporal patterns of mangrove change and the spatial variability of environmental drivers.

1.2 Significance of the Study

Mangrove forests are a productive ecosystem that offers various ecosystem services that benefit all humankind. Not only does it provide raw materials, and protection in coastal areas, but it also plays a crucial role in mitigating the increasing climate change impact. With mangrove's sequestration ability, it became part of the "blue carbon" ecosystems along with other coastal wetlands namely seagrasses and tidal marshes. These blue carbon ecosystems are highly efficient in storing significant amounts of carbon for a long period of time (Choudhary et al., 2024), making them effective and efficient in mitigating the impact of climate change. According to Donato et al. (2011), mangrove is one of the most carbon-rich forests. Hence, safeguarding this productive ecosystem not only contributes to protection of coastal areas but also helps combat the increasing impact of climate change. Overall, the findings of this study are expected to help conservationists and policy makers in making evidence-based and scientific-based conservation strategies and policies for the protection of the mangrove ecosystem in the Philippines.

1.3 Research Questions

This study aims to answer the following questions:

- What are the spatial and temporal patterns of mangrove cover change in Palawan between 1996, 2010, 2015, and 2020?
- Which environmental variables are significantly associated with mangrove cover change?
- How do the effects of these environmental variables vary across different locations in Palawan?

1.4 Research Objectives

The objective of this study is to analyze the spatio-temporal patterns of mangrove loss in the Philippines from 2000 to 2020 and identify the key environmental drivers influencing this change. In particular, this study aims to:

1. Map and quantify spatial and temporal patterns of mangrove cover change in Palawan, Philippines, by comparing mangrove extent across the years 1996, 2010, 2015, and 2020 using Global Mangrove Watch data.
2. Analyze the relationship between mangrove change and environmental drivers using logistic regression.
3. Evaluate spatial variability in driver influence through Geographically Weighted Regression (GWR).

1.5 Related Literature

The Philippines, as an archipelagic country with more than 7,000 islands, has one of the longest coastlines in the world, extending up to 36,289 kilometers. This makes it home to

various coastal ecosystems, including mangroves. As estimated by Long et al. (2011), in the year 2000, the Philippines had approximately 256,185 hectares of mangrove forests, with Palawan having the largest area. However, despite its geographic location that allows various species of mangroves to thrive, mangrove cover continues to decline at an alarming rate due to both natural and anthropogenic disturbances. According to Long et al. (2014), there was a 10.5% decrease in the total area of mangroves in the Philippines from 1990 to 2010, with Zamboanga Sibugay, Palawan, Zamboanga del Sur, Bohol, and Negros Occidental showing the greatest areal loss. As the country is continuously exposed to both natural and human pressures, the status of mangrove ecosystems also constantly changes. Hence, monitoring the spatial distribution of mangrove ecosystems is necessary to identify areas most in need of immediate protection and rehabilitation. However, observing large extents of mangrove forests using traditional methods is time-consuming, costly, and nearly impossible for inaccessible sites. With recent technological developments, ecological trends and historical changes in mangrove forests, including those in remote areas, can now be monitored using remote sensing at a much faster rate and lower cost. This enables the acquisition of accurate, timely, and reliable data on mangrove forests. Remote sensing has been widely used for mapping and monitoring threatened mangrove ecosystems (Kuenzer et al., 2011). The availability of remotely sensed data, improved image pre-processing, and skilled users have enabled mangrove monitoring at both local and global scales (Giri, 2016). Global Mangrove Watch (GMW), a remote sensing platform, was developed to address the need for timely mangrove information on a global scale (Bunting et al., 2018). This platform provides users with tools to observe historical changes in mangrove forests. According to Ochiai et al. (2022), this tool can be used by countries lacking in-house monitoring systems to assess progress toward their mangrove-related commitments. Hence, by offering up-to-date information on mangrove extent, GMW assists conservationists and researchers in monitoring mangrove forests at local, regional, and global levels.

Mangrove degradation is often attributed to anthropogenic drivers such as aquaculture expansion, urbanization, and overexploitation of natural resources (Alongi, 2002; Polidoro et al., 2014; Quevedo et al., 2022). In the Philippines, aquaculture development—particularly for fish and shrimp—is a well-documented cause of mangrove conversion and habitat loss (Long et al., 2011). While such transformations are frequently justified in the name of food security, they result in the destruction of critical habitats and the loss of ecosystem services (Tengku et al., 2021). However, mangrove loss is also increasingly driven by environmental stressors, especially in the context of the global climate crisis. The Philippines is among the most vulnerable nations to tropical cyclones, which generate storm surges capable of uprooting mangrove stands and devastating entire forest patches (Buitre et al., 2019). Rising sea levels and altered inundation regimes further stress mangrove systems, particularly seaward zones. Prolonged submergence can lead to tree mortality due to oxygen deprivation and salinity shifts, as noted by Ward et al. (2016). These physical drivers—though less often quantified in spatial studies—are ecologically significant and deserve closer scrutiny. This study, therefore, emphasizes the role of common environmental variables such as rainfall, temperature, and elevation in influencing mangrove change. Given the lack of reliable, spatially aligned data on anthropogenic pressures at the scale of analysis, the research pivots toward assessing how natural physical gradients contribute to the spatial variability in mangrove dynamics in Palawan.

To appropriately develop a strategic plan for mangrove conservation and protection, it is essential to apply analytical tools that examine the relationships between environmental drivers and their influence on mangrove extent. Various statistical modeling techniques can be employed for this purpose. Logistic regression, a statistical method used when the dependent variable is binary (e.g., presence or absence of mangrove loss), allows researchers to assess how factors such as rainfall, temperature, and elevation affect the likelihood of mangrove degradation. Identifying which disturbances significantly influence mangroves can inform targeted interventions. However, the effects of these drivers often vary across space. Geographically Weighted Regression (GWR), a spatial analysis technique, captures this variation by estimating location-specific relationships between predictors and mangrove loss (Nazeer & Bilal, 2018). Together, logistic regression and GWR enable the assessment of both global and localized drivers of mangrove change, providing a robust evidence base for spatially informed conservation strategies.

Although spatial regression techniques have been applied in various global contexts, their integration remains limited in Philippine mangrove research. This study addresses a gap in existing research by applying spatial regression models to mangrove data in a Philippine context, offering localized insights into mangrove loss drivers.

2. Methodology

2.1 Study Area

The study was conducted in the province of Palawan, Philippines (Figure 1), known for its extensive and diverse mangrove ecosystems. Composed of 23 municipalities and one (1) highly urbanized city (HUC), Palawan lies between approximately 7.50°N to 12.67°N latitude and 116.93°E to 121.33°E longitude with a total land area of approximately 17,000 square kilometers. It encompasses complex coastal landscapes and island environments. Its geographical and ecological characteristics make it an ideal region for analyzing mangrove dynamics and their environmental driver.

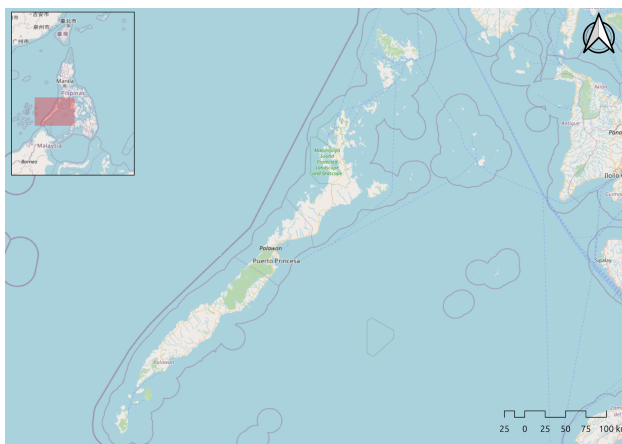


Figure 1. Map of the study area

The entire methodological workflow of the research is shown in Figure 2.

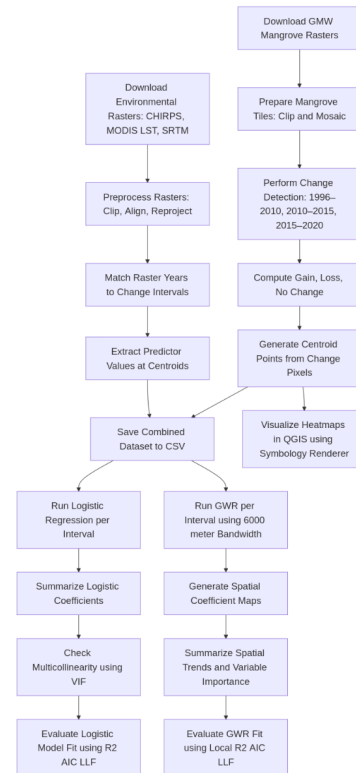


Figure 2. Flowchart of the research methodology

2.2 Data Acquisition

This study relied entirely on open-access geospatial datasets covering the years 1996, 2010, 2015, and 2020. These temporal checkpoints were chosen based on the availability of mangrove extent data and consistent multi-source environmental datasets. All datasets were selected based on spatial resolution, temporal alignment, and relevance to mangrove ecosystems and their drivers.

2.2.1 Mangrove Cover

Mangrove extent data were obtained from the Global Mangrove Watch (GMW) Version 3.0. The dataset was developed by the Japan Aerospace Exploration Agency (JAXA) and its partners. The GMW dataset maps the geographical distribution of mangrove forests at a spatial resolution of 25 meters, and is distributed as raster GeoTIFFs in the WGS 84 geographic coordinate system (EPSG:4326).

The dataset was derived using ALOS PALSAR synthetic aperture radar (SAR) backscatter and other Earth Observation archives. For this study, four temporal snapshots were used: 1996, 2010, 2015, and 2020. These years were selected to capture long-term as well as decadal changes in mangrove extent. The rasters represent binary presence-absence values:

- Pixel value 1 indicates the presence of mangrove
- Pixel value 0 (or NoData) indicates absence or non-mangrove areas

The GMW tiles covering the Philippine archipelago were first downloaded and then clipped to the spatial extent of Palawan province using a vector boundary shapefile. These rasters form the foundation for the change detection component of the study.

2.2.2 Explanatory Variables

To identify potential biogeographical and environmental drivers of mangrove change, the study incorporated three explanatory variables: rainfall, land surface temperature (LST), and elevation. These variables were selected based on their documented ecological relevance to mangrove distribution and health, such as rainfall dependency, temperature tolerance, and elevation-based zonation. Additionally, they were available as spatial raster datasets with consistent formats and resolutions for the selected years of analysis: 2010, 2015, and 2020. Table 1 summarizes the information about the datasets.

The CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data) product was used to capture annual rainfall patterns, combining satellite imagery with station-based observations at ~5 km spatial resolution. The MODIS Land Surface Temperature (MOD11A2 v6.1) dataset provides 8-day composites of temperature at 1 km resolution, which were aggregated to produce annual averages for each study year. SRTM elevation data at 30-meter resolution were used as a static topographic layer to account for elevation-based constraints on mangrove establishment and persistence, particularly in relation to tidal inundation thresholds and coastal slope.

Table 1. Summary of explanatory variables

Variable	Source	Spatial Resolution	Temporal Availability	Type
Rainfall	CHIRPS	~5 km	2010, 2015, 2020	Raster
Land Surface Temperature	MODIS (8-day average)	1 km	2010, 2015, 2020	Raster
Elevation	SRTM	30 m	Static	Raster

2.3 Data Preprocessing

2.3.1 Mangrove Raster Preprocessing

The Global Mangrove Watch (GMW) raster datasets used in this study were originally provided as tiled GeoTIFF files covering the global tropics. To isolate the study area, a GeoJSON boundary file representing the extent of Palawan was used to clip and extract only those tiles overlapping with the province. The tiles for the years 1996, 2010, 2015, and 2020 were first identified and filtered spatially, then mosaicked together using Python's rasterio and geopandas libraries.

After mosaicking, each annual raster was clipped to the exact extent of the Palawan boundary. The resulting raster layers represented binary mangrove presence (value = 1) and absence (value = 0) at 25-meter resolution. To ensure consistency in analysis, all rasters were confirmed to share the same coordinate reference system (EPSG:4326) and grid alignment. This preprocessed set of rasters served as the basis for subsequent change detection analysis.

2.3.2 Explanatory Variable Preprocessing

Each environmental explanatory raster—rainfall (CHIRPS), land surface temperature (MODIS), and elevation (SRTM)—was preprocessed using a unified pipeline to ensure consistency in spatial extent, resolution, and coordinate reference system.

Global-scale datasets were first clipped to the extent of Palawan using a common GeoJSON boundary consistent with that used for the mangrove extent layers. CHIRPS rainfall data were obtained as annual mean precipitation rasters for 2010, 2015, and 2020 at ~5 km resolution and required only clipping and alignment. MODIS Land Surface Temperature (MOD11A2 v6.1), which provides 8-day composites at 1 km resolution, was processed in Google Earth Engine (GEE). For each year of interest, MODIS images were filtered by date, masked to valid values, and reduced to annual means before being exported as GeoTIFF files.

The SRTM elevation layer, which is static and available at 30-meter resolution, was merged from multiple tiles and clipped to the study area. As elevation is temporally invariant, only spatial alignment and CRS matching were necessary.

After clipping, all rasters were reprojected to a common coordinate reference system (WGS 84, EPSG:4326), facilitating pixel-wise comparison and regression analysis across all explanatory layers.

2.3.3 Temporal and Spatial Harmonization

To ensure analytical consistency across datasets, all mangrove and environmental explanatory layers were temporally and spatially harmonized. For each of the three mangrove change intervals—1996–2010, 2010–2015, and 2015–2020—the corresponding set of environmental predictors was selected from the nearest available year. Specifically, mangrove loss between 1996 and 2010 was linked to predictors from 2010; loss between 2010 and 2015 used predictors from 2015; and loss from 2015 to 2020 used predictors from 2020. This temporal alignment assumes that landscape and climatic conditions at the end of each interval are most representative of the drivers influencing observed change.

Instead of using raster overlay, a point-based sampling approach was implemented. Centroid points were generated for each pixel classified as mangrove loss, and these points were then used to extract values from the explanatory rasters. This approach provides spatial precision and ensures compatibility among datasets with differing native resolutions. Harmonization at both temporal and spatial levels allowed for consistent sampling of environmental conditions associated with each loss event, enabling statistically robust regression and geospatial analyses.

2.4 Change Detection Analysis

Mangrove change detection was conducted by comparing binary raster layers from the Global Mangrove Watch (GMW) dataset across three temporal intervals: 1996–2010, 2010–2015, and 2015–2020. Each raster indicates mangrove presence (1) or absence (0) at a 25-meter resolution. For each interval, change was identified through pairwise raster differencing, where the earlier year's raster was subtracted from the later year's raster. This operation resulted in new change rasters with the following classification:

- +1 indicates gain (new mangroves appeared where there were none)
- -1 indicates loss (mangroves disappeared)
- 0 indicates no change
- NoData was assigned to areas where both rasters had no valid data

To support spatial interpretation, centroid points were generated for all gain and loss pixels. These were used to visualize the spatial distribution of change using heatmap rendering techniques, highlighting clusters or hotspots of mangrove dynamics.

For each time interval, the number of pixels in each change category was calculated and converted to hectares (1 pixel = 625 m² or 0.0625 ha). Spatial summaries of mangrove loss and gain were also produced at the municipal level, enabling sub-provincial analysis of mangrove change across Palawan.

2.5 Regression Modeling

To identify the key environmental drivers of mangrove loss, the study employed a two-stage regression modeling approach: (1) global logistic regression, and (2) localized modeling using Geographically Weighted Regression (GWR). This framework aimed to quantify both the strength and direction of variable influence, as well as the spatial variation in these relationships across Palawan.

The analysis focused exclusively on pixels identified as mangrove loss, as determined by the change detection analysis. For each of the three temporal intervals—1996–2010, 2010–2015, and 2015–2020—a corresponding dataset was compiled containing the geographic coordinates of each loss pixel, along with its extracted values from the environmental raster layers: rainfall (CHIRPS), land surface temperature (MODIS LST), and elevation (SRTM). This produced three independent datasets aligned with the temporal windows of mangrove change.

Before analysis, pixels with missing or invalid data were excluded. Additionally, continuous predictor variables were normalized to account for differences in measurement scale (e.g., mm of rainfall, meters of elevation, °C temperature). Logistic regression was performed using the statsmodels Python library, while GWR was conducted using the mgwr package to explore spatial heterogeneity in model coefficients.

2.5.1 Logistic Regression

For each mangrove change interval, a logistic regression model was fitted using a binary dependent variable indicating the presence (1) or absence (0) of mangrove loss. The model included three continuous environmental predictors: rainfall, land surface temperature (LST), and elevation. The logistic model estimates the probability of mangrove loss as a function of these variables:

$$\text{logit}(p) = \beta_0 + \beta_1 \cdot \text{Rainfall} + \beta_2 \cdot \text{LST} + \beta_3 \cdot \text{Elevation}$$

Separate models were run for each explanatory year (2010, 2015, and 2020), corresponding to the three change intervals. For each model, summary statistics including coefficient estimates, standard errors, p-values, and pseudo R² values were extracted. Model fit was evaluated using metrics such as log-likelihood, Akaike Information Criterion (AIC), and the likelihood ratio chi-squared statistic.

To assess the relative importance of each predictor, standardized coefficient magnitudes and their statistical significance were examined. Multicollinearity was evaluated using a correlation matrix and Variance Inflation Factors (VIF), which indicated acceptable levels of collinearity across all models.

2.5.2 Geographically Weighted Regression (GWR)

While logistic regression provides a global model of variable influence, it assumes that the relationships between predictors and mangrove loss are spatially uniform. To examine how these relationships vary across space, Geographically Weighted Regression (GWR) was applied using the same set of environmental predictors: rainfall, land surface temperature (LST), and elevation. GWR estimates localized regression coefficients for each sample point by fitting separate models that incorporate only nearby data, weighted by geographic proximity.

A fixed bandwidth of 6,000 meters was selected to balance spatial sensitivity with the resolution of the input variables. This choice was guided primarily by the CHIRPS rainfall dataset, which has a spatial resolution of approximately 5 km. A bandwidth smaller than this would fail to capture meaningful variation in rainfall across neighborhoods, since many sample points would share identical values. The 6 km bandwidth ensures sufficient differentiation among rainfall values, while still allowing the model to detect localized variation in LST (1 km) and elevation (30 m), which operate at finer spatial scales.

The GWR analysis produced spatially explicit coefficient layers for each predictor, illustrating how the strength and direction of their influence varied across Palawan. These were visualized as maps to reveal regional patterns. Additionally, model fit metrics such as local pseudo R², Akaike Information Criterion (AIC), and log-likelihood were computed for performance assessment. Finally, variable importance was ranked per year by summarizing the spatial distribution of coefficients.

3. Results and Discussion

3.1 Mangrove Extent from 1996 to 2020

Figure 3 illustrates the temporal trajectory of mangrove extent in Palawan across four benchmark years. In 1996, mangroves occupied approximately 55,968 hectares, forming extensive coastal fringes along bays, estuaries, and river mouths—key ecotones in tropical island biogeography. By 2010, this extent had declined by over 1,400 hectares, marking a significant contraction in the spatial footprint of these intertidal forests. This contraction is likely linked to a combination of anthropogenic encroachment and shifting hydrogeomorphic conditions.

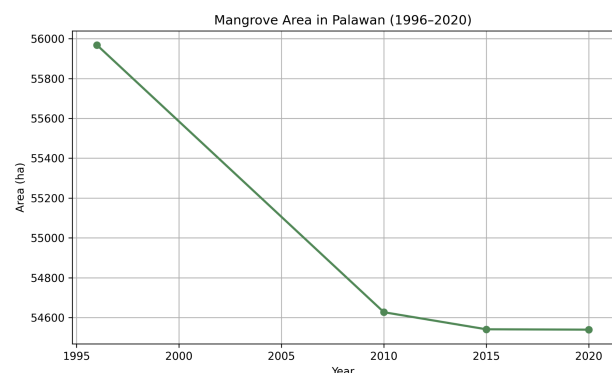


Figure 3. Mangrove Extent in Palawan from 1996 to 2020

In the subsequent decade, the rate of decline decelerated, with only marginal reductions observed between 2015 and 2020. This trend may signal an emerging equilibrium phase in certain

zones, where biotic resilience, regeneration capacity, or policy-driven protection (e.g., buffer zoning or community reforestation) are beginning to counterbalance long-term pressures. Nonetheless, the cumulative decline over 24 years underscores the continuing vulnerability of Palawan's mangrove ecosystems—particularly in low-lying deltas and urban-adjacent coasts where disturbance regimes are spatially concentrated.

3.2 Mangrove Change Detection

Mangrove change across Palawan was evaluated over three ecologically significant intervals: 1996–2010, 2010–2015, and 2015–2020, based on differencing of binary presence-absence rasters from the Global Mangrove Watch dataset. Table 3 summarizes the spatial extent of each change category per interval in hectares.

Table 3. Total Mangrove Change in Palawan by Time Interval

Interval	Loss (ha)	Gain (ha)	No Change (ha)
1996–2010	2,013	672	53,955
2010–2015	515	429	54,112
2015–2020	1,056	1,054	53,485

The interval 1996–2010 exhibited the most extensive mangrove contraction, with over 2,000 hectares lost and minimal recovery (672 hectares gained), indicative of widespread disturbance or conversion along critical coastal fringes and estuarine deltas. In contrast, the 2010–2015 period marked a shift in mangrove dynamics, with a significant reduction in net loss and a narrowing gap between loss and gain—a potential signal of local ecosystem resilience or intervention-driven stability.

By 2015–2020, the system approached net equilibrium, with gains (1,054 ha) nearly matching losses (1,056 ha). This emerging balance suggests possible biogeographic compensation in certain coastal segments, where historical degradation may now be paralleled by natural regeneration or targeted rehabilitation. Notably, over 53,000 hectares of mangroves remained consistently unchanged across all intervals, highlighting the spatial persistence of core mangrove habitats in more geophysically stable or less-accessible zones.

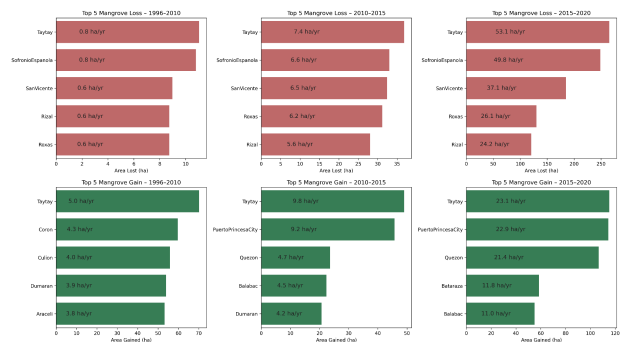


Figure 4. Top 5 Municipalities by Mangrove Loss and Gain per Time Interval with Annual Rates

At the municipality level, spatial and temporal dynamics of mangrove change across Palawan's municipalities reveal important patterns that complement the provincial-scale analysis. As illustrated in Figure 4, the top five municipalities with the highest mangrove loss and gain are shown for each

time interval, along with their average annual rates of change in hectares per year.

A notable contrast emerges when comparing the earliest interval (1996–2010) to the more recent ones. While this period registered the highest total mangrove loss at the provincial level (over 2,000 hectares), the annual rates of change within individual municipalities were relatively low, averaging under 1 ha/year for loss and under 5 ha/year for gain. This suggests that early mangrove decline was spatially diffuse, likely reflecting widespread but moderate pressures distributed across many coastal settlements and estuarine zones. However, it is also important to consider that the 1996–2010 interval spanned 14 years, nearly three times longer than the two succeeding 5-year intervals. The larger total loss in this period may thus represent cumulative degradation over a longer time frame rather than higher yearly pressures. This temporal disparity underscores the need to analyze both absolute change and rate of change when evaluating ecosystem dynamics.

In the 2010–2015 and especially the 2015–2020 intervals, mangrove change became significantly more spatially concentrated and ecologically intense. Municipalities such as Taytay, Sofronio Española, and San Vicente showed dramatic surges in loss rates, exceeding 35 ha/year, while others like Puerto Princesa City, Quezon, and Balabac led in gains. Notably, Taytay recorded the highest rate of both loss (53.1 ha/year) and gain (23.1 ha/year) in the most recent interval, indicating high landscape turnover—possibly due to simultaneous development pressures and restoration efforts in different barangays or coastal zones.

These evolving patterns suggest a broader biogeographical shift: from an earlier phase of low-intensity, widespread contraction to a contemporary regime of localized but rapid change. Such a transition may be driven by a combination of policy interventions, population redistribution, aquaculture expansion, and natural regeneration. This highlights the increasing spatial heterogeneity of mangrove dynamics in Palawan and reinforce the need for municipality-specific conservation strategies that can respond to both ecological vulnerability and recovery potential.



Figure 5. Heatmaps of mangrove loss (top row) and gain (bottom row) in Palawan across three time intervals (1996–2010, 2010–2015, and 2015–2020)

In addition to quantitative summaries, the spatial distribution of mangrove loss and gain was visualized using heatmaps derived from change centroids (Figure 5). These maps reveal localized concentrations of change that are not immediately apparent

from previous visualizations. For example, the southern and central municipalities exhibit persistent hotspots of loss across all intervals, while gain patterns are more scattered, episodic, and scarce. The spatial overlap of loss and gain in areas such as Taytay and Puerto Princesa suggests high landscape turnover—reflecting simultaneous degradation and restoration activities. The fine-scale nature of these patterns highlights the importance of sub-municipal monitoring and targeted conservation strategies. Furthermore, the visualization reinforces earlier findings of increasing spatial intensity of mangrove dynamics in the later intervals.

3.3 Environmental Drivers of Mangrove Loss (Logistic Regression)

To assess the biogeographic correlates of mangrove loss, logistic regression was performed for each change interval using normalized environmental predictors: rainfall (CHIRPS), land surface temperature (MODIS LST), and elevation (SRTM). The regression models estimate the probability of mangrove loss (vs. no loss) based on these environmental variables. Table 4 summarizes the model coefficients and significance values across the three time slices. All predictors were statistically significant at the 0.001 level, but their direction and magnitude of influence varied across years.

Table 4. Logistic Regression Coefficients and Model Fit per Interval

Predictor	2010 Coef	2015 Coef	2020 Coef
Rainfall (CHIRPS)	−0.0002	−0.4433	+0.1446
Temperature (LST)	+0.0027	−0.1772	−0.4971
Elevation	−0.2774	−2.1652	−1.8971
Pseudo R ²	0.1228	0.0431	0.0354

2010 (Loss from 1996–2010)

In this period, higher land surface temperatures and elevation were positively associated with mangrove loss, while rainfall had a relatively weak negative effect. The model explained approximately 12% of the variance in mangrove loss, suggesting that environmental conditions in 2010 were reasonably predictive of the preceding decade’s changes.

2015 (Loss from 2010–2015)

The model for 2015 showed the strongest negative association between elevation and mangrove loss (coef = −2.1652), implying that lower-lying mangrove areas were more susceptible to degradation. Both rainfall and temperature were also negatively associated with loss. The pseudo R² was 0.043, indicating a modest fit but statistically significant explanatory power.

2020 (Loss from 2015–2020)

In contrast to 2015, higher rainfall in 2020 was positively associated with loss, suggesting potential impacts from extreme precipitation or storm events in this interval. Temperature and elevation maintained their negative associations with loss, with elevation again emerging as the strongest predictor. The pseudo

R² was 0.035, indicating consistent, albeit moderate, predictive strength.

Interpretation and Biogeographic Implications

The consistent negative influence of elevation across all models reflects the ecological constraints of mangrove zonation—these ecosystems are adapted to tidal zones and thus restricted to low-lying areas. Higher elevation areas are less likely to support mangroves, and loss events occurring in low-elevation areas likely reflect either natural disturbance or anthropogenic conversion.

The role of temperature (LST) fluctuates across intervals, but its overall negative association in later years may indicate that cooler microclimates better support mangrove persistence, or that higher surface temperatures exacerbate drought stress or soil salinization in marginal mangrove areas.

The behavior of rainfall is more complex: its negative association in 2015 suggests water availability supports mangrove health, while the positive coefficient in 2020 could point to extreme rainfall or typhoon-driven stressors as drivers of recent loss.

Collectively, these patterns underscore the environmental sensitivity of mangrove systems in Palawan. While spatially consistent predictors such as elevation serve as a baseline constraint, climatic variability—especially temperature and precipitation—plays a fluctuating role in modulating mangrove vulnerability over time.

These findings align with core principles of biogeography, particularly the concept of environmental filtering. Elevation acts as a stable biogeographic filter that governs the fundamental spatial limits of mangrove ecosystems. Low-elevation coastal zones offer favorable conditions for mangrove establishment due to tidal inundation and salinity gradients—but they are also the most exposed to anthropogenic disturbances such as aquaculture, development, or altered hydrology. The persistence of mangrove loss in these areas likely reflects the convergence of ecological sensitivity and spatially concentrated human pressures. This intersection creates distinct biogeographic vulnerability zones—areas where physical suitability is high but ecological resilience is low due to overlapping disturbance regimes.

3.4 Spatial Drivers of Mangrove Loss: GWR Results, Model Evaluation, and Diagnostics

3.4.1 GWR Coefficient Patterns

Geographically Weighted Regression (GWR) revealed important spatial variation in how environmental variables influence mangrove loss across Palawan. The model estimated localized coefficients for each variable—elevation, land surface temperature (LST), and rainfall—allowing an interpretation of how their effects shifted not only over time, but also across different biophysical zones.

Across all three predictor years (2010, 2015, and 2020), elevation consistently showed a strong negative relationship with mangrove loss, with mean coefficients of −0.175 in 2010, −0.062 in 2015, and −0.074 in 2020. These values suggest that lower elevation zones are more vulnerable to mangrove degradation, likely due to their increased exposure to tidal inundation, saltwater intrusion, and development pressure along

accessible low-lying coasts. The weakening influence in later years could imply that human activity has begun to affect mangroves even in areas beyond the traditional intertidal zone.

Land Surface Temperature (LST) was also negatively associated with loss, particularly in 2010 (mean = -0.013) and 2020 (-0.008). This pattern suggests that warmer mangrove zones—possibly associated with open canopies, exposed shorelines, or deforested areas—may be more prone to degradation. In 2015, however, the relationship flattened (mean ≈ -0.001), which could reflect anomalous climate conditions or weaker spatial gradients in LST influence that year.

Rainfall (CHIRPS) showed the weakest and most inconsistent relationship with mangrove loss. Mean coefficients hovered near zero across all years, though a slight negative shift was observed in 2020 (-0.007). This suggests that rainfall is not a spatially consistent predictor of mangrove loss, though in specific sites, high rainfall may still contribute to erosion, flooding, or sedimentation stress.

These patterns are summarized in Figure 6, which shows the average GWR coefficient per variable over time. Elevation's declining yet dominant role is clearly visible, while LST and rainfall fluctuate with weaker and more temporally variable influence.

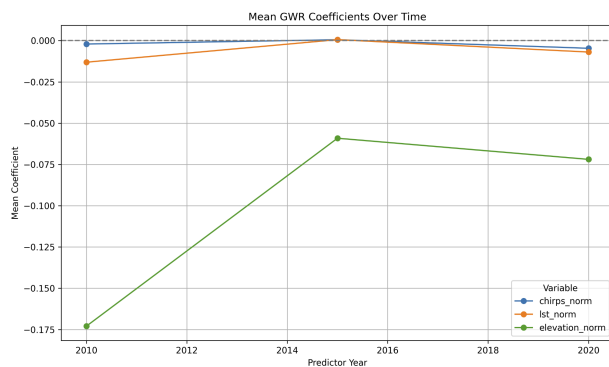


Figure 6. Mean GWR coefficients for each predictor from 2010 to 2020

3.4.2 Spatial Interpretation of GWR Coefficients

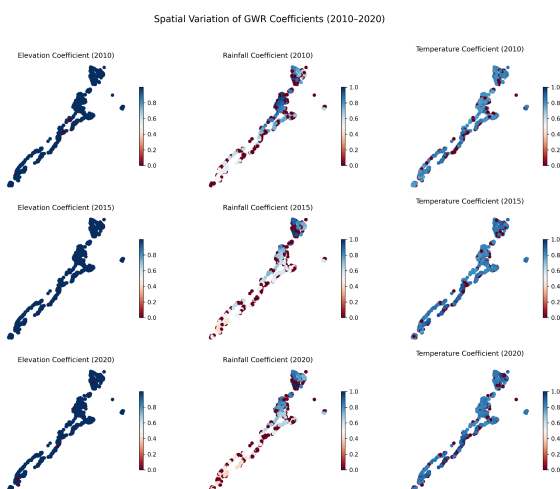


Figure 7. Spatial variation of GWR coefficients for elevation, rainfall, and temperature across Palawan (2010–2020)

The spatial distribution of GWR coefficients offers deeper insight into how environmental drivers of mangrove loss vary geographically across Palawan. Figure 7 presents the spatialized coefficients for elevation, rainfall, and land surface temperature (LST) for the years 2010, 2015, and 2020.

Elevation

This spatial consistency affirms elevation's role as a biogeographic filter—a stable environmental gradient that governs mangrove zonation. Areas closest to sea level align with the species' physiological tolerances for tidal inundation and salinity, while areas even slightly higher in elevation exceed those limits, leading to natural boundaries of mangrove distribution. What GWR further reveals is that the degree of sensitivity to this filter is not uniform: in central and southern municipalities, these low-elevation zones not only coincide with suitable habitat but also with regions experiencing intensified human activity. This intersection of ecological suitability and anthropogenic exposure represents a biogeographic vulnerability hotspot.

Rainfall

In contrast, rainfall (CHIRPS) shows substantial spatial heterogeneity. In 2010 and 2015, some areas such as northern Palawan and the west coast display slightly positive coefficients (reddish hues), suggesting that in these zones, higher rainfall may be contributing to mangrove degradation—potentially due to flooding, sedimentation, or storm-related disturbance. However, much of the province exhibits weak or neutral coefficients, indicating that rainfall is a highly localized and context-dependent driver. By 2020, this pattern becomes even more fragmented, possibly due to interactions between rainfall, land cover change, and erosion patterns. These localized differences point to the interaction between microclimate variation and biogeographic context—where the same climatic variable may have opposite effects depending on coastal morphology, exposure to storms, and the surrounding land use configuration.

Land Surface Temperature (LST)

Temperature shows moderate to strong spatial signals across all years. In 2010, northern and interior coastal areas show stronger negative coefficients, while by 2015 and 2020, the negative influence extends across more of the central belt. This may reflect zones where warmer surface conditions—linked to canopy loss, edge exposure, or reduced shading—increase mangrove vulnerability. The pattern suggests that LST is not just a passive climatic variable, but a possible indicator of thermal or anthropogenic stress gradients in coastal areas.

3.4.3 GWR Model Fit and Predictive Strength

The performance of each GWR model was evaluated using three global metrics: pseudo R^2 , Akaike Information Criterion (AIC), and residual sum of squares (RSS), as shown on Table 5. These indicators assess how well the model explains variation in mangrove loss while penalizing complexity and overfitting.

Table 5. Global GWR model performance metrics across predictor years

Year	Pseudo R ²	AIC	RSS (Residual Sum of Squares)
2010	0.034	93,587.63	62,060.84
2015	0.071	34,085.24	17,052.44
2020	0.058	60,743.32	31,689.14

The highest pseudo R² was observed in 2015 (0.071), indicating the model explained approximately 7% of the variation in mangrove loss using only spatialized environmental predictors. Though the values may appear modest, this level of explanatory power is not uncommon in ecological regression models with binary outcomes, particularly at fine spatial scales where unmeasured anthropogenic or geomorphological factors may dominate.

AIC values were lowest in 2015, suggesting that this model achieved the best balance of goodness-of-fit and model parsimony. The lower RSS in 2015 further confirms that residual error was smallest during this time slice, reinforcing the interpretive strength of environmental variables for explaining mangrove loss in this interval.

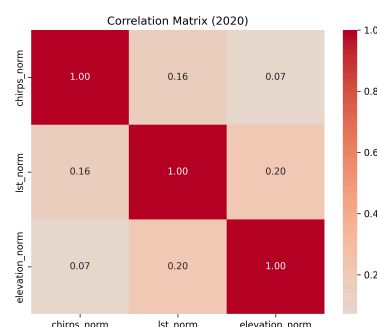
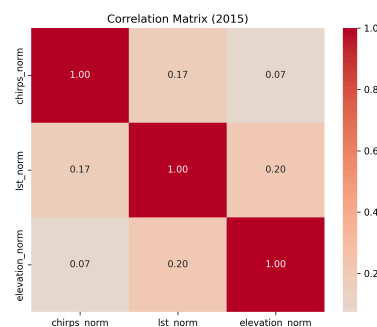
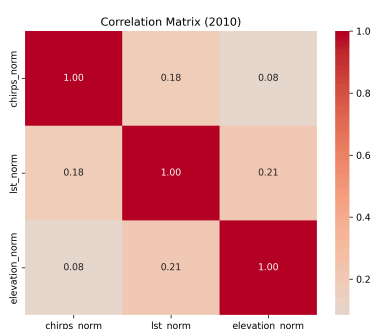
The slightly lower pseudo R² in 2020 (0.058) and 2010 (0.034) may reflect increased noise in the spatial data or the influence of external, non-environmental drivers such as land tenure shifts, aquaculture conversion, or urban expansion.

3.4.4 Multicollinearity Diagnostics

To evaluate the robustness of the regression models, multicollinearity diagnostics were conducted after model fitting. While the regression analyses (logistic regression and GWR) had already provided insights into the influence of explanatory variables, it was important to confirm that the predictors used were not excessively correlated, which could distort coefficient estimates and model interpretation.

Pairwise Correlation Matrices

As visualized in Figures 8a–8c, the Pearson correlation coefficients among rainfall (CHIRPS), land surface temperature (LST), and elevation were consistently low to moderate across all three years. In 2010, the highest correlation was only 0.21 (between LST and elevation), with other pairs showing even weaker relationships. Similar patterns were observed in 2015 and 2020, with no correlation exceeding 0.22.



Figures 8a–8c. Correlation matrices for normalized predictors in 2010, 2015, and 2020

Variance Inflation Factor (VIF)

VIF scores offer a more formal diagnostic of multicollinearity by quantifying how much the variance of each regression coefficient is inflated due to linear dependence among predictors. All VIF values in this study were well below the conservative threshold of 5.0, with most scores close to 1.00 (see Table 6).

Table 6. Variance Inflation Factors (VIF) for normalized predictors by year

Year	Rainfall VIF	LST VIF	Elevation VIF
2010	1.04	1.05	1.01
2015	1.03	1.05	1.01
2020	1.02	1.05	1.01

These diagnostics reinforce the interpretability of the regression outputs, suggesting that the observed variable effects are statistically reliable and not inflated due to inter-predictor redundancy.

4. Conclusion

This study analyzed the spatial and temporal dynamics of mangrove loss in Palawan, Philippines, over a 24-year period (1996–2020) using a combination of remote sensing change detection, regression modeling, and spatial analysis. Results revealed that while the most substantial mangrove decline occurred between 1996 and 2010, the rate of loss slowed in the succeeding years, with gains nearly matching losses by 2020. This shift in trend may signal localized regeneration, conservation success, or changing land-use pressures.

Regression analysis showed that elevation and rainfall were consistently significant drivers of mangrove loss across all time intervals. Geographically Weighted Regression (GWR) further demonstrated the spatial non-stationarity of these effects, with environmental influence varying markedly across coastal regions. The integration of pixel-level environmental data and local regression models provided biogeographically meaningful insights, affirming the environmental sensitivity of mangrove systems and the complex interplay between stable topographic constraints and temporally fluctuating climatic drivers across island landscapes like Palawan. In sum, the observed spatial patterns are not merely statistical outputs—they reflect underlying biogeographic processes such as tidal zonation, ecological edge vulnerability, and filtering by elevation and exposure. Recognizing these ecological foundations is key to designing conservation strategies that are attuned to both the physical geography and the biotic constraints of mangrove ecosystems.

Given the observed trends, future research should expand the inclusion of environmental factors such as soil salinity, tidal regimes, and coastal hydrodynamics to capture the nuanced controls on mangrove viability. Additionally, incorporating higher-resolution anthropogenic indicators—such as aquaculture footprints, road expansion, or zoning data—could enhance understanding of local pressures. From a methodological standpoint, future work should explore temporally dynamic models (e.g., panel or time-series GWR) and improve integration with socioecological frameworks. Continuous monitoring through updated satellite datasets and community-based validation is also recommended to support adaptive coastal management and policy implementation.

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